CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 5: Factorisation

Topic-wise short notes with solved examples

Methods of factorising algebraic expressions: common factors, grouping, splitting the middle term, difference of two squares, perfect squares, and the sum and difference of two cubes.

1 What factorising means

- To **factorise** an expression is to write it as a product of simpler expressions (its factors). It is the reverse of expanding.
- Always look for a **common factor first**. Then choose a method from the number of terms.

Number of terms	Try
Two	Difference of squares; sum or difference of cubes
Three	Perfect square; splitting the middle term
Four or more	Grouping; a perfect square with one extra term

- Factorise **completely**: keep going until no factor can be broken further.

2 Common factors

- Take out the HCF of all the terms. The common factor may itself be a bracket.

Example 2.1 Factorise $12a^2b - 18ab^2 + 6ab$.

SOLUTION

$6ab(2a - 3b + 1)$

Example 2.2 Factorise $x(a - b) - y(b - a)$.

SOLUTION

$b - a = -(a - b)$, so the expression is $x(a - b) + y(a - b)$.

$(a - b)(x + y)$

3 Grouping

- Arrange the terms in groups so that each group has a common factor and the brackets left behind are the same.

Example 3.1 Factorise $ax + by + ay + bx$.

SOLUTION

$(ax + ay) + (bx + by) = a(x + y) + b(x + y)$

$(x + y)(a + b)$

Example 3.2 Factorise $a^2 - ab - 3a + 3b$.

SOLUTION

$a(a - b) - 3(a - b)$

$(a - b)(a - 3)$

Example 3.3 Factorise $x^2 + \frac{1}{x^2} + 2 - 5x - \frac{5}{x}$.

SOLUTION

$\left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right)$

$\left(x + \frac{1}{x}\right)\left(x + \frac{1}{x} - 5\right)$

4 Trinomials: splitting the middle term

- For $x^2 + bx + c$: find two numbers with **product** c and **sum** b.
- For $ax^2 + bx + c$: find two numbers with **product** ac and **sum** b, split the middle term, then group.
- If the product is positive the two numbers have the same sign as b. If it is negative they have opposite signs.
- If $b^2 - 4ac$ is not a perfect square, the trinomial has no factors with rational coefficients.

Example 4.1 Factorise $x^2 + 7x + 12$.

SOLUTION

Product 12, sum 7: the numbers are 3 and 4.

$(x + 3)(x + 4)$

Example 4.2 Factorise $x^2 - 5x - 24$.

SOLUTION

Product -24, sum -5: the numbers are -8 and 3.

$(x - 8)(x + 3)$

Example 4.3 Factorise $6x^2 + 11x - 10$.

SOLUTION

Product -60, sum 11: the numbers are 15 and -4.

$6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5)$

$(2x + 5)(3x - 2)$

Example 4.4 Factorise $2(a + b)^2 - 5(a + b) - 3$.

SOLUTION

Let $y = a + b$: $2y^2 - 5y - 3 = (2y + 1)(y - 3)$

$(2a + 2b + 1)(a + b - 3)$

5 Difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

- A **sum** of two squares $a^2 + b^2$ has no real factors.
- Perfect-square trinomials: $a^2 + 2ab + b^2 = (a + b)^2$ and $a^2 - 2ab + b^2 = (a - b)^2$.
- Four terms may form a perfect square and one more square: group three and one.

Example 5.1 Factorise $25x^2 - 49y^2$.

SOLUTION

$(5x + 7y)(5x - 7y)$

Example 5.2 Factorise $a^4 - 16$.

SOLUTION

$(a^2 + 4)(a^2 - 4)$

$(a^2 + 4)(a + 2)(a - 2)$

Example 5.3 Factorise $x^2 - 6x + 9 - 4y^2$.

SOLUTION

$(x - 3)^2 - (2y)^2$

$(x - 3 + 2y)(x - 3 - 2y)$

Example 5.4 Factorise $x^4 + x^2 + 1$.

SOLUTION

Add and subtract x^2: $(x^2+1)^2 - x^2$

$(x^2 + x + 1)(x^2 - x + 1)$

Example 5.5 Evaluate $(6.7)^2 - (3.3)^2$.

SOLUTION

$(6.7 + 3.3)(6.7 - 3.3) = 10 \times 3.4$

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6 Sum and difference of two cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2), \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

- The sign in the first bracket matches the sign in the expression. The middle sign in the second bracket is the opposite. The last sign is always plus.

Example 6.1 Factorise $8x^3 + 27y^3$.

SOLUTION

$(2x)^3 + (3y)^3$

$(2x + 3y)(4x^2 - 6xy + 9y^2)$

Example 6.2 Factorise $a^3 - \frac{1}{a^3}$.

SOLUTION

$\left(a - \frac{1}{a}\right)\left(a^2 + 1 + \frac{1}{a^2}\right)$

Example 6.3 Factorise $a^6 - b^6$.

SOLUTION

$(a^3 + b^3)(a^3 - b^3)$

$(a + b)(a - b)(a^2 - ab + b^2)(a^2 + ab + b^2)$

Example 6.4 Factorise $2x^4 + 16x$.

SOLUTION

$2x(x^3 + 8)$

$2x(x + 2)(x^2 - 2x + 4)$

7 Common mistakes

- Stopping early: $a^4 - 16 = (a^2 + 4)(a^2 - 4)$ is not complete.
- Trying to factorise $a^2 + b^2$.
- Sign errors when a group starts with a minus: $-3a + 3b = -3(a - b)$.
- For $ax^2 + bx + c$, using numbers whose product is c instead of ac.
- Writing $a^3 + b^3 = (a + b)(a^2 + ab + b^2)$. The middle term is $-ab$.
- Not taking out the common factor first.

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